

Rapport

Purpose

The purpose of this analysis is to examine which factors influence insurance costs.

Method

The analysis was performed using data cleaning, descriptive statistics, and visualization. A regression model was used to analyze the relationship between insurance costs and selected variables.

Result

The results show that insurance costs differ between groups. Smokers, older individuals, and people with higher bmi tend to have higher insurance costs. Differences between genders and regions are relatively small.

Data understanding

The dataset was loaded into R using the `read_csv()` function. It contains 1100 observations and 14 variables.

The dataset includes both *numerical variables* (age, bmi, children, prior_accidents, prior_claims, annual_checkups, charges) and *categorical variables* (customer_id, sex, region, smoker, chronic_condition, exercise_level, plan_type).

Missing values were checked using `colSums(is.na())`. The variables with missing values are:

- bmi (28 missing values) => numerical variable
- exercise_level (22 missing values) => categorical variable
- annual_checkups (20 missing values) => numerical variable

Missing values will be handled during the data cleaning step.

Data cleaning and preparation

```
> unique(df_raw$sex)
[1] "female" "male"
> unique(df_raw$region)
[1] "east" "west" "south" "north" "North" "South"
> unique(df_raw$smoker)
[1] "no" "yes" "Yes"
> unique(df_raw$chronic_condition)
[1] "no" "yes"
> unique(df_raw$exercise_level)
[1] "high" "medium" "low" NA
> unique(df_raw$plan_type)
[1] "basic" "standard" "premium" "Premium" "Standard"
```

Missing values in numerical variables (bmi and annual checkups) were replaced with the median, while missing values in categorical variables (exercise level) were replaced with "Unknown". Text variables were also standardized by converting values to title case.

A new variable *age_group* was created that helps categorize customers into different life stages: young adult (18–29), adult (30–49), middle-aged (50–64) and senior (65+). Most individuals are in the adult group (615), followed by middle-aged (262), young adult (171), and senior (52). This grouping makes it easier to compare insurance costs between different age groups.

Descriptive analysis

The descriptive analysis shows how insurance costs are distributed across groups.

The Adult group has the highest total insurance costs (6096881) and the largest number of customers (615), followed by the Middle-aged group (2876276) and customers (262). The Senior group has the lowest total costs (651505), and also includes the fewest customers (52). In the Adult and Middle-aged groups, some individuals have much higher costs than others.

The analysis also shows that insurance costs differ by region. The South region has the highest total insurance costs (3297950) and the largest number of customers (320), while the East region has the lowest total costs (2493169) and the fewest customers (253).

The distribution of insurance costs by smoking status, gender, and region shows that smoking has a strong effect on insurance costs. Smokers have significantly higher costs than non-smokers, while differences between males and females are very small, as are regional differences.

Regression analysis

The predictors were selected based on the descriptive analysis. Smoking status and age were identified as the most important variables, as they showed clear differences in insurance costs. Bmi was included because it is related to health and may affect insurance costs. Gender and children were also included as control variables.

The results of the regression model show that smoking has the strongest effect on insurance costs. Being a smoker increases insurance costs by approximately 7968 units. Age and bmi also affect insurance costs. Older people and people with higher bmi usually pay more for insurance. Gender and children do not have a clear effect on insurance costs.

The results seem reasonable, as smoking, age and bmi are related to health risks, which can increase insurance costs.

The analysis is based on a limited dataset, which may not represent all cases. Some missing values were replaced during data cleaning, which may affect the results. Also, not all factors

that may influence insurance costs are included in the model, for example lifestyle, health history, income, etc.

Interpretation and conclusions

Strengths:

The model includes relevant variables such as smoking, age and bmi. The results are easy to interpret.

The factors that have the strongest relationship with insurance costs are smoking, age, and bmi.

The analysis shows that smokers, older persons, and people with higher bmi tend to have higher insurance costs.

The model does not include all factors that may affect costs, such as lifestyle, health history, or income.

The model could be improved by including more variables and using more advanced methods.

Self-reflection

1.What do you think you did well in the assignment?

I think I did well in cleaning the data, creating clear visualizations, and explaining the results in a simple and understandable way.

2.What was the most difficult?

The most difficult part was understanding the regression model and interpreting the results correctly.

3.What grade do you think your submission corresponds to, G or VG?

I did my work at a G level.

4.Justify briefly by referring to which requirements in the assignment you think you met.

I have completed the following parts: 1) Data understanding, 2) data cleaning and preparation, 3) descriptive analysis, 4) regression analysis, and 5) interpretation and conclusions.